Skills Summary

Distributions, Models, and Qualified Inference

Use this reference while completing your worksheet or when studying.

Skill 1 — Summarising a distribution

Two summaries, two jobs. The **mean** uses every value and moves when an extreme value does; the **median** uses position and does not.

Median of an even-sized ordered list = the mean of the two middle values. **Quartiles** (school convention): split the ordered list into halves and take the median of each; IQR = $Q_3 - Q_1$.

Reading shape from the pair: mean above median $\Rightarrow$ right-skewed; mean below median $\Rightarrow$ left-skewed; roughly equal $\Rightarrow$ roughly symmetric.

Example: Find the median of $4,\ 7,\ 9,\ 12,\ 15,\ 20$.
Step 1. The list is even-sized, so no single value is in the middle — the median is defined as the mean of the two central values, and the list is already ordered.
Step 2. The two central values are 9 and 12, so the median is $(9+12)\div 2$. $\Rightarrow$ median = **10.5**

Try it: Find the median of $3,\ 5,\ 8,\ 11,\ 14,\ 20$.

check: **9.5**

Skill 2 — Comparing two models with residuals

Residual = observed − predicted, always in that order.

- **Sign** says direction: positive $\Rightarrow$ the model under-predicted there; negative $\Rightarrow$ it over-predicted.
- **Size** says accuracy. Compare two models by comparing |residual|, or by summing squared residuals over all points.
- **Pattern** says whether the model has the right *shape*. Signs scattered at random $\Rightarrow$ right shape. All one sign, or a run of signs that flips across the range $\Rightarrow$ wrong shape, however small the misses are.

Example: A model predicts $y = 5x + 3$. At $x = 4$ the observed value is 22. Find the residual and read it.
Step 1. Comparing a model to data means residuals, so evaluate the model at that input and subtract it from the observed value in that order.
Step 2. Predicted $= 5(4) + 3$, so the residual is $22 - (5(4) + 3) = -1$, and the negative sign says the observation fell below the model. $\Rightarrow$ residual = $\mathbf{-1}$, the model over-predicted

Try it: A model predicts $y = 6x + 4$. At $x = 3$ the observed value is 19. Find the residual.

check: $\mathbf{-3}$

Skill 3 — Qualifying an inference

Before stating a conclusion, answer all four:

- **Range.** Is the input inside the observed data? Inside is interpolation; outside is an assumption.
- **Plausibility.** Does the situation itself limit the output — a percentage above 100, a culture that must run out of food?
- **Bias.** Do the residuals lean one way? A model can be accurate and still systematically low.
- **Sample.** How many independent runs or subjects? One run gives no variability, so no margin of error can be attached.

Probability from counts: $P = \dfrac{\text{favourable}}{\text{total}}$, in lowest terms. Count carefully: a residual of exactly zero is neither positive nor negative.

Example: Of 5 residuals, 3 are positive. One of the five points is chosen at random. Find the probability it has a positive residual.
Step 1. All points are equally likely, so the probability is the count of favourable points over the total count — no model arithmetic is needed.
Step 2. Three of the five qualify, so $P = 3 \div 5$, already in lowest terms. $\Rightarrow \quad P = \mathbf{\frac{3}{5}}$

Try it: Of 8 residuals, 2 are positive. One of the eight points is chosen at random. Find the probability it has a positive residual.

check: $\frac{1}{4}$